

FORT SEVERN, ON, Canada

WMO: 710990

Lat: 56.020N

Lon: 87.680W

Elev: 52

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.0	(c) -30.0	(d) -41.6	(e) 0.5	(f) -34.0	(g) -37.5	(h) 0.6	(i) -29.8	(j) 23.3	(k) -3.0	(l) 21.9	(m) -4.7	(n) 7.2	(o) 250	(p) 0.867

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	80.5	64.9	75.6	62.7	71.0	60.7	66.9	76.1	64.0	73.1	61.7	69.5	13.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	86.3	71.6	60.4	78.5	68.2	58.0	71.9	65.6	31.6	75.7	29.3	73.2	27.7	69.3	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.1	20.6	18.5	-40.6	88.7	5.6	4.5	-44.7	92.0	-47.9	94.6	-51.1	97.1	-55.2	100.3		
			-41.5	71.5	5.7	2.8	-45.6	73.5	-49.0	75.2	-52.2	76.8	-56.4	78.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.7	-9.6	-8.4	3.6	18.6	34.4	46.2	54.9	55.2	47.3	35.6	17.3	-1.1
	DBStd	25.63	12.50	10.90	13.92	12.28	8.99	9.19	7.89	7.00	7.27	7.06	12.46	15.53
	HDD50	9753	1849	1634	1439	943	495	183	34	21	139	451	982	1583
	HDD65	14755	2314	2054	1904	1393	949	570	329	318	532	912	1432	2048
	CDD50	512	0	0	0	0	10	70	187	183	58	4	0	0
	CDD65	40	0	0	0	0	0	7	17	15	1	0	0	0
	CDH74	564	0	0	0	0	7	139	256	147	14	1	0	0
	CDH80	142	0	0	0	0	0	37	71	33	1	0	0	0
Wind	WSAvg	9.6	8.9	9.0	9.5	9.7	10.2	10.0	9.1	9.3	9.9	10.1	10.0	9.4
Precipitation	PrecAvg	21.6	0.7	0.6	0.8	0.9	1.8	2.4	3.2	3.1	2.9	2.4	1.6	1.0
	PrecMax	26.2	1.5	1.6	1.7	2.3	3.9	4.2	6.3	7.0	6.7	5.8	3.0	2.7
	PrecMin	14.4	0.2	0.1	0.0	0.3	0.6	0.9	1.3	1.5	1.1	0.7	0.7	0.2
	PrecStd	3.0	0.3	0.4	0.4	0.5	0.9	0.8	1.2	1.1	1.3	1.1	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.6	27.0	44.7	51.1	73.9	84.7	86.7	84.4	75.9	62.2	44.0	31.7
		MCWB	28.5	25.3	39.2	42.2	56.7	64.6	68.2	69.5	63.2	54.2	39.7	30.8
	2%	DB	23.0	19.9	38.8	45.7	63.7	78.0	81.1	77.8	71.0	55.4	38.0	28.6
		MCWB	22.5	18.2	34.5	38.4	51.7	61.5	65.0	64.3	61.1	50.1	35.7	28.0
	5%	DB	17.4	14.1	32.6	41.3	54.8	70.4	75.6	73.3	64.5	49.4	34.7	25.7
		MCWB	16.8	12.6	29.8	35.8	46.5	58.3	63.4	63.0	57.2	46.2	33.3	25.0
	10%	DB	10.5	8.6	25.5	36.6	47.7	63.7	70.4	67.8	59.6	44.7	32.5	21.9
		MCWB	9.6	7.3	23.4	33.0	42.6	55.8	61.0	60.6	53.8	42.3	31.5	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.9	25.5	39.1	43.3	56.9	68.8	70.8	70.1	65.2	56.3	40.4	31.4
		MCDB	29.7	26.7	43.8	50.5	71.8	79.1	81.2	82.1	72.9	62.2	42.9	32.3
	2%	WB	22.5	18.3	34.6	38.7	51.8	62.9	67.1	66.5	61.2	50.8	36.3	28.3
		MCDB	22.9	19.8	38.2	44.4	63.1	74.6	77.2	74.3	69.1	55.4	37.9	28.7
	5%	WB	16.8	12.5	29.7	36.3	47.1	59.5	64.1	63.7	57.6	46.3	33.6	25.5
		MCDB	17.3	14.0	33.0	41.0	54.0	69.7	73.7	72.0	64.3	49.8	34.7	26.0
	10%	WB	9.8	7.4	23.5	32.9	42.9	55.1	61.7	61.1	54.2	42.5	31.8	21.7
		MCDB	10.4	8.5	25.6	36.6	48.2	64.2	69.3	67.4	59.1	44.5	32.7	22.4
Mean Daily Temperature Range	5% DB	MDBR	16.0	18.6	21.6	18.0	15.0	19.5	20.9	17.1	15.0	10.0	12.8	13.7
		MCDBR	19.3	23.1	24.5	22.8	30.3	31.8	30.4	26.0	24.5	17.6	11.3	11.3
		MCWBR	19.2	21.7	21.0	16.7	19.4	17.6	17.2	14.5	15.7	13.1	9.9	11.3
	5% WB	MCDBR	19.3	23.2	23.5	21.4	29.6	30.2	27.3	23.2	22.8	16.8	10.8	11.0
		MCWBR	19.2	21.9	20.3	16.2	19.0	18.8	17.1	14.0	15.7	12.9	9.7	11.0
Clear-Sky Solar Irradiance	taub	0.234	0.257	0.282	0.360	0.382	0.372	0.401	0.373	0.339	0.302	0.259	0.216	
	taud	2.134	2.173	2.167	2.005	2.109	2.274	2.316	2.418	2.501	2.522	2.288	2.126	
	Ebn at Noon	217	259	280	269	271	276	264	266	261	246	208	195	
	Edh at Noon	18	26	34	47	46	39	37	31	25	19	16	14	
All-Sky Solar Radiation	RadAvg	263	605	1110	1617	1725	1889	1708	1307	843	438	193	161	
	RadStd	23	46	75	132	157	181	151	116	88	39	33	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-74	N/A

Nomenclature: See separate page